

LD9-V0 Mesh Fidelity and Topology Validation Specification

Sampled surface distance, normal error, scalar residual, and indexed-mesh topology

mdstats development specification

2026-07-21

Contents

LD9-V0 - Mesh Fidelity and Topology Metrics	1
Purpose	1
Input constraints	1
Topology summary	1
Surface sampling	2
Symmetric sampled distance	2
Normal error	2
Scientific scalar residual	2
Fidelity policy	2
Schemas	2
Focused tests	2
Limitations and next stage	3

LD9-V0 - Mesh Fidelity and Topology Metrics

Package target: mdstats 0.19.54a0

Status: implemented calibration metrics; simplification is not implemented here

Module: mdstats.plotting.density_mesh_validation

Purpose

This module defines deterministic validation metrics for comparing a candidate browser mesh with a raw reference mesh or, when supplied, the underlying scientific scalar field.

The module does not modify either mesh and does not perform mesh simplification.

Input constraints

Vertices are finite arrays with shape $(n_vertices, 3)$. Faces are integer arrays with shape $(n_faces, 3)$. Every face index must be valid, and a face may not repeat a vertex index. Zero-area triangles are rejected for geometric comparison.

Topology summary

```
MeshTopologySummary(  
    vertex_count,  
    edge_count,  
    face_count,  
    connected_component_count,  
    boundary_edge_count,  
    nonmanifold_edge_count,  
    euler_characteristic,  
)
```

Undirected edge incidence is counted from indexed triangles. The Euler characteristic is

$$\chi = V - E + F.$$

A mesh is reported as a closed two-manifold when every used edge has incidence two.

This summary is a validation diagnostic. Periodic seam pairing remains a separate LD9-V1/V2 requirement.

Surface sampling

Triangles are sampled proportionally to area. For a selected triangle with vertices ($\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$), two deterministic pseudorandom variates are reflected into the unit simplex and the point is

$$\mathbf{x} = \mathbf{a} + u(\mathbf{b} - \mathbf{a}) + v(\mathbf{c} - \mathbf{a}).$$

The seed and maximum sample count are explicit. Identical meshes sampled with the same seed produce identical point sets.

Symmetric sampled distance

Let P and Q be the sampled reference and candidate point sets. The sampled symmetric distance collection is

$$D(P, Q) = \left\{ \min_{q \in Q} \|p - q\| : p \in P \right\} \cup \left\{ \min_{p \in P} \|q - p\| : q \in Q \right\}.$$

The report stores median, 99th percentile, and maximum values. This is a bounded calibration approximation to symmetric surface distance; later production validation may add adaptive refinement near high-error regions.

Normal error

Triangle normals accompany every sampled point. Each sample is paired with its nearest opposite-mesh sample. Orientation-independent angular error is

$$\theta = \cos^{-1}(|\mathbf{n}_1 \cdot \mathbf{n}_2|).$$

The report stores median, 99th percentile, and maximum angles in degrees.

Scientific scalar residual

When a scalar sampler and contour level ρ_q are supplied, candidate samples are checked using

$$r(\mathbf{x}) = |\rho(\mathbf{x}) - \rho_q|.$$

The sampler must return one finite value per candidate sample. Scalar residual validation is optional in V0 but becomes required for LD9-V2 acceptance against the scientific field.

Fidelity policy

```
MeshFidelityOptions(  
    max_samples=50_000,  
    random_seed=0,  
    max_surface_error=0.02,  
    max_normal_error_degrees=8.0,  
    max_scalar_residual=None,  
    require_component_count=True,  
    require_euler_characteristic=True,  
    require_closed_two_manifold_match=True,  
)
```

The report is passing only when every enabled requirement passes.

Schemas

```
mdstats.mesh-topology-summary.v1  
mdstats.mesh-fidelity-options.v1  
mdstats.mesh-fidelity-report.v1
```

Focused tests

Tests must include:

- a closed octahedron with $V = 6$, $E = 12$, $F = 8$, and $\chi = 2$;
- identical-mesh zero-distance comparison;
- translated-mesh surface-error rejection with preserved topology;
- scalar-residual evaluation against an explicit contour function;
- malformed face and zero-area rejection;

- canonical JSON round trips for topology, options, and fidelity reports.

Limitations and next stage

The V0 nearest-sample metric is suitable for calibration and regression tests but is not a formal exact Hausdorff computation. LD9-V2 should supplement it with adaptive field-based sampling, periodic seam checks, and protected-component validation.